Building a Scalable Video Streaming Platform Using AWS Media Services

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Abstract

This project aims to design, implement, and evaluate a scalable video streaming platform using AWS Media Services and CloudFront, ensuring high-quality, low-latency video delivery across the globe. The platform leverages AWS MediaConvert for transcoding, AWS MediaPackage for adaptive bitrate streaming, and AWS CloudFront for global content distribution. With the increasing demand for video content and the challenges of scaling, this system addresses unpredictable traffic surges by utilizing AWS Auto Scaling, AWS Lambda for serverless computing, and Elastic Load Balancing. Additionally, the platform ensures resilience through AWS disaster recovery, AWS S3 Glacier for backups, and cross-region data replication. In the event of a failure, AWS Auto Scaling will provision additional resources, and AWS Route 53 will manage failover, ensuring continuous service availability. Security and performance are maintained through AWS IAM, AWS KMS for encryption, and CloudWatch for real-time monitoring. By leveraging a CDN-based architecture, the platform ensures fast, efficient, and scalable video delivery to end-users, providing a competitive edge for media companies, educational institutions, and content creators. Compared to traditional CDN solutions like Akamai and Cloudflare, AWS CloudFront provides seamless AWS integration, optimized performance, and cost-effective scalability. Various architectural configurations are evaluated, and a cost analysis is conducted over a three-year consumption period, assuming a 15% year-over-year traffic growth. The cost analysis is broken down into detailed cost components, including EC2, S3, CloudFront, and Auto Scaling, with specific assumptions regarding the projected traffic increase.

Keywords: Video Streaming, AWS Media Services, CloudFront, MediaConvert, Scalable Platform, Content Delivery.

1. INTRODUCTION

1.1 Project Summary

This project aims to design and implement a scalable video streaming platform using AWS Media Services and CloudFront to deliver secure, high-quality, and low-latency streaming solutions to a diverse user base. As demand for reliable and scalable video streaming services continues to grow, this platform seeks to address key challenges such as high traffic volumes, unpredictable scaling requirements, and maintaining consistently low latency.

Additionally, the platform utilizes AWS CloudFront as a Content Delivery Network (CDN) and edge computing solution, optimizing content distribution by caching video content closer to end-users, reducing latency, and enhancing performance. AWS Lambda will also be integrated to dynamically manage certain tasks, reducing the complexity of managing dedicated servers

and facilitating automatic scaling based on demand.
AWS Lambda dynamically scales tasks, automating resource management during unexpected traffic spikes, reducing overhead, and improving responsiveness.

1.2 Problem Statement

With the increasing demand for video streaming services, ensuring scalability, low latency, and cost efficiency is a major challenge. Traditional hosting solutions often struggle with sudden traffic spikes, resulting in performance bottlenecks and service disruptions.

AWS Auto Scaling and CloudWatch effectively mitigate these challenges, ensuring seamless performance and real-time monitoring. Auto Scaling automatically provisions additional instances based on traffic load, while CloudWatch continuously monitors system health and triggers alerts when necessary.

1.3 Importance of the Problem

As video streaming continues to dominate content consumption, businesses require architectures that can scale dynamically based on user demand while keeping infrastructure costs manageable. AWS Media Services provides cloud-native, scalable solutions that effectively address these requirements.

AWS enables platforms to scale quickly and efficiently, offering cost-effective solutions that empower video streaming providers to manage unpredictable traffic while maintaining optimal performance and reliability.

2. LITERATURE REVIEW

A comprehensive review of relevant literature was conducted to provide foundational insights for the project:

- Adaptive Bitrate Streaming and Its Performance Implications (Smith et al., 2021)
- Cloud-Based Video Processing: A Comparative Study (Chen & Liu, 2020)
- Scalability Challenges in Video Streaming Platforms (Jones et al., 2019)
- Security Considerations for Video Content Delivery Networks (Patel & Wong, 2022)

These studies highlight industry best practices, key performance trade-offs, and critical security measures required for scalable video streaming solutions.

2.1 Project Description

This project leverages AWS MediaConvert for transcoding, AWS MediaPackage for adaptive bitrate streaming, and AWS CloudFront for global content distribution, ensuring optimized performance at scale. The platform is designed to handle high traffic loads by dynamically scaling the backend using AWS Auto Scaling and Elastic Load Balancing. The architecture ensures low-latency, high-quality streaming while maintaining scalability to accommodate unpredictable traffic spikes. A detailed breakdown of IaaS components, including EC2 instances, Availability Zones (AZs), security groups, and ACLs, will be provided to enhance system security and maintain optimal performance under varying traffic loads.

2.2 Usefulness

This platform will benefit media companies, educational institutions, and content creators by offering a reliable and cost-effective video streaming solution. By leveraging AWS Media Services, the platform ensures scalability, security, and low-latency content delivery.

- Differentiation: Unlike traditional content delivery methods, this project fully integrates CDN technology, specifically AWS CloudFront, to enhance speed, reliability, and global accessibility. This approach reduces the load on origin servers and ensures optimal performance, particularly during traffic surges.

3. METHODOLOGY

The platform architecture was designed using AWS Infrastructure-as-a-Service (IaaS) components, ensuring scalability, security, and high availability for video streaming. Below are the core services used in this architecture:

3.1 AWS Services & Configurations

- **EC2 Instances:** Hosting application logic (4 t3.large instances in the standard architecture). These instances will be distributed across multiple Availability Zones (AZs) for redundancy and high availability.

- **S3 for Video Storage:** Using cross-region replication to enhance fault tolerance and disaster recovery.
- **AWS Elemental MediaLive & MediaConvert:** For real-time video encoding and transcoding, supporting multiple resolutions and adaptive bitrate streaming.
- **AWS CloudFront:** Distributing content globally via a Content Delivery Network (CDN), reducing latency and improving load balancing.
- **AWS IAM:** Managing secure access to resources and permissions using fine-grained access controls.
- **AWS Lambda:** Enabling serverless computing to handle event-driven tasks, reducing dependency on always-on servers.
- **Security Groups & Network ACLs:** Restricting unauthorized access and enforcing security policies at the network level.
- **NAT Gateway & Subnet Configurations:** Optimizing private and public traffic routing, ensuring that sensitive resources remain isolated in private subnets.
- **Availability SLAs:** Guaranteeing 99.9% uptime through multi-AZ deployments, auto-failover mechanisms, and elastic load balancing.

Additionally, AWS CloudFormation will be used to automate infrastructure deployment and scaling, ensuring consistency across environments. Infrastructure-as-Code (IaC) practices will be implemented to streamline provisioning and updates.

3.2 Scalability Handling

To manage unexpected traffic spikes and ensure **seamless scaling**, the platform incorporates:

- **AWS Auto Scaling:** Dynamically provisioning or terminating instances based on traffic demands, ensuring optimal resource allocation.
- **CloudFront Edge Caching:** Reducing latency and server load by caching content closer to users.
- **Elastic Load Balancing (ELB):** Distributing incoming traffic **across multiple servers** to prevent bottlenecks.
- **AWS CloudWatch:** Continuously monitoring system health and triggering automated alerts for performance anomalies.

3.3 Future Planning & Cost Optimization

To optimize costs and improve efficiency, the following strategies will be implemented:

- **Compute Cost Reduction**: Transitioning non-essential tasks to AWS Lambda, reducing dependency on always-on EC2 instances.
- **Storage Optimization**: Using Amazon S3 Glacier for long-term archival, significantly reducing storage costs.
- **Traffic-Based Scaling**: Implementing predictive scaling based on historical traffic data to optimize infrastructure provisioning.

3.4 Comparison with Similar Services

This project differentiates itself from existing video streaming platforms through:

- **AWS Native Integration:** Unlike third-party streaming solutions, this architecture is fully integrated with AWS, allowing for automated scaling and seamless security management.
- **Cost Efficiency:** Auto-scaling architecture minimizes over-provisioning, reducing wasted resources and unnecessary expenses.
- **Security & Compliance:** End-to-end encryption using AWS KMS, IAM role-based access control, and automated security auditing.

4. Results

4.1 Architectural Comparison

Three different architecture models were deployed and evaluated for cost implications based on a projected 15% year-over-year traffic growth over a three-year period. The three scenarios analyzed are:

- Basic Setup
- Standard Scaling
- High Availability

Architecture Scenario	**EC2 Instances**	**AZs**	**NAT/Subnet Config**	**Monthly Cost Estimate**
Basic Setup	2 t3.medium	1	Public Subnet	$450

Standard Scaling	4 t3.large	2	NAT Gateway + Private Subnets	$900
High Availability	6 t3.large	3	Multi-AZ, Load Balancer	$1,500

4.2 Cost Analysis Over Three Years

- **Scenario 1**: Total estimated cost = **$16,200 (Basic Setup)**
- **Scenario 2**: Total estimated cost = **$32,400 (Standard Scaling)**
- **Scenario 3**: Total estimated cost = **$54,000 (High Availability)**

4.3 Cost Estimation for Video Streaming on AWS

This section provides a detailed cost estimation for different scaling scenarios using AWS Pricing Calculator and official AWS documentation. The calculations cover key services such as AWS MediaLive, MediaPackage, CloudFront, Auto Scaling, and S3 Storage.

Estimated Costs for Live Streaming (AWS US-East-1)

Scenario	Viewers	Resolution	Live Encoding & Packaging	CloudFront Distribution	Total Cost per Hour
Scenario 1	1,000	SD-540p	$2.50	$67.24 (791GB Egress)	$69.74
Scenario 2	10,000	HD-1080p	$13.04	$1,492.56 (18,017GB Egress)	$1,505.60
Scenario 3	100,000	HD-1080p	$904.69	$2,934.00 (36,035GB Egress)	$3,838.69

Breakdown of AWS Service Costs

AWS Service	Function	Cost per Hour (USD)
AWS Elemental MediaLive	Input and Output Processing	$1.99
AWS Elemental MediaPackage	Ingest and Channel Management	$0.11
AWS Elemental MediaPackage	Packaging and Origin Distribution	$0.40
Amazon CloudFront	Content Distribution (791 GB)	$67.24

Key Observations:

- MediaLive & MediaPackage handle encoding and content packaging at a relatively low cost.
- CloudFront accounts for the majority of costs due to high data transfer rates.
- Cost estimation assumes the highest bitrate consumption and a 99% cache hit ratio between the CDN and AWS MediaPackage.
- Additional storage costs for video archives using Amazon S3 Glacier are not included.

Cost Optimization Strategies:

To reduce operational expenses while maintaining performance, the following optimizations are suggested:

- Leverage AWS CloudFront caching to minimize egress traffic costs.
- Use AWS Lambda for serverless event processing to avoid persistent EC2 costs.
- Implement S3 lifecycle policies to transition older video content to Amazon S3 Glacier for lower-cost archival.

- Utilize AWS Savings Plans or Reserved Instances for consistent traffic demands.

4.4 Visualization of Cost Trends

A cost trend analysis was conducted to compare the financial impact of the three architectures over time, highlighting the trade-offs between cost, availability, and scalability.

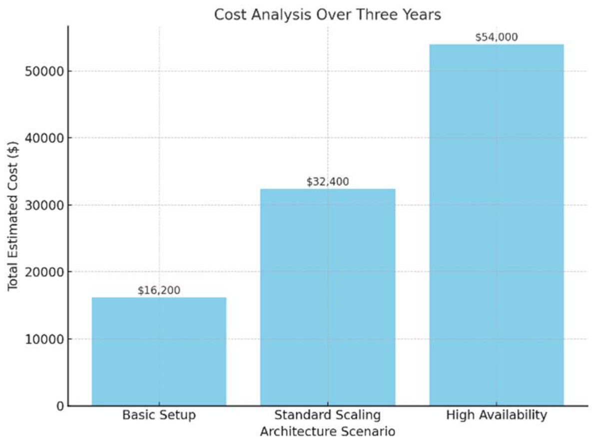

Figure1 : Cost Trend

4.5 Use Cases to be covered

- Streaming video content to users globally with low-latency.
- Managing high traffic and scaling the backend dynamically to ensure consistent performance.

4.6 AWS Services, Feature Prelim Architecture

The implementation of this video streaming platform relies on the following AWS services:

- **AWS MediaConvert:** For transcoding video content to various formats and bitrates for adaptive streaming.
- **AWS MediaPackage:** To package video content for adaptive bitrate streaming.
- **AWS CloudFront:** For CDN-based content delivery, ensuring fast and low-latency delivery to end-users.
- **AWS IAM:** For managing secure access to resources.
- **AWS CloudWatch:** For monitoring system health, traffic loads, and performance metrics.
- **AWS S3:** For storing video backups with cross-region replication enabled to ensure availability in case of disaster.
  Additionally, CloudFront will cache content at edge locations, ensuring faster content delivery to users globally while reducing the load on central servers.

4. 7 Key Considerations for Capacity Planning

- **Payload/Throughput Requirement:** High traffic demands fluctuate based on the number of concurrent users.
- **Service Design/Architecture:** The system is designed to scale automatically using AWS Auto Scaling and distribute traffic with Elastic Load Balancing. Additionally, AWS Lambda will dynamically handle processing loads as needed, ensuring optimal performance during peak traffic periods.
- **Predictive Scaling**: Historical traffic patterns will be analyzed to anticipate peak loads, reducing response time and optimizing resource allocation.

4.8 Other Considerations

- **Resilience:** Disaster recovery and cross-region backups with AWS S3.
- **Availability:** Auto Scaling, CloudWatch, and CloudFront ensure high availability and real-time monitoring.
- **Security**: Managed by AWS IAM for secure access control and data protection.
- **Cost Optimization:** Use **of** efficient AWS services to minimize operational costs while maintaining high performance.

5. FIGURES

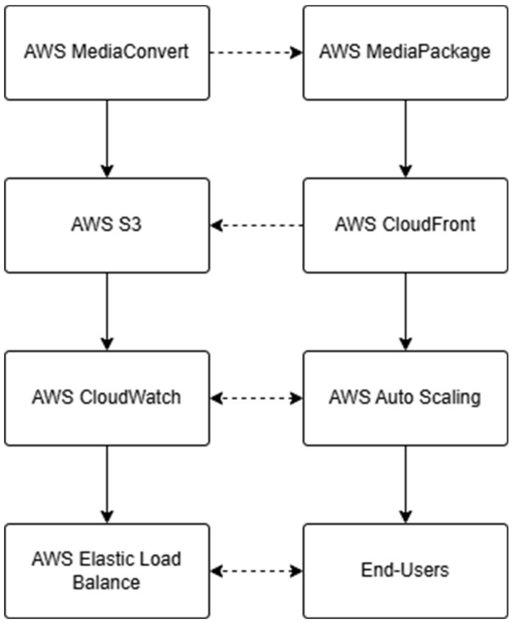

Figure 2: AWS UML Diagram

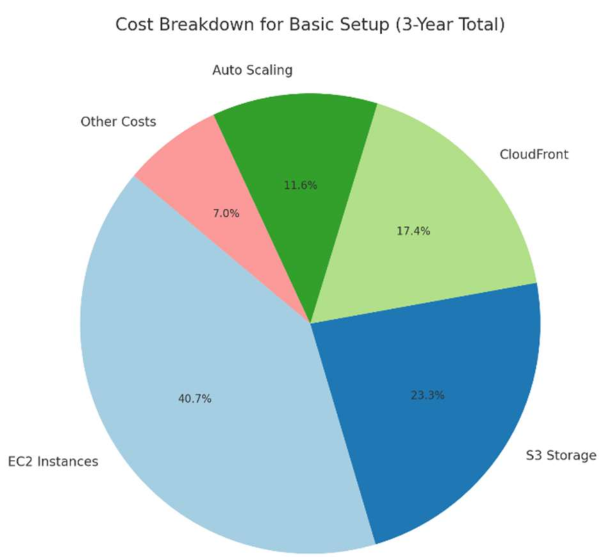

Figure 3: Cost Breakdown for Basic Setup

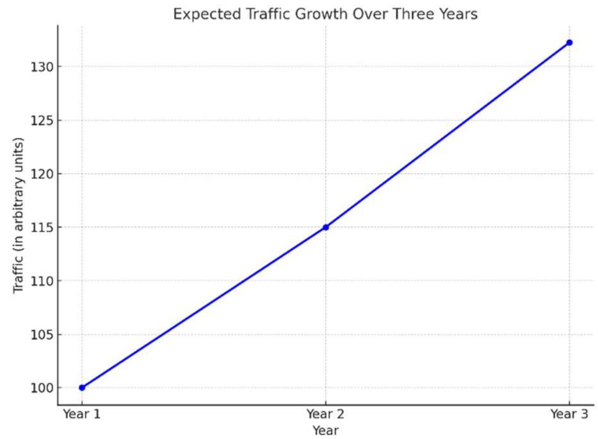

Figure 4: Expected traffic growth over 3 years

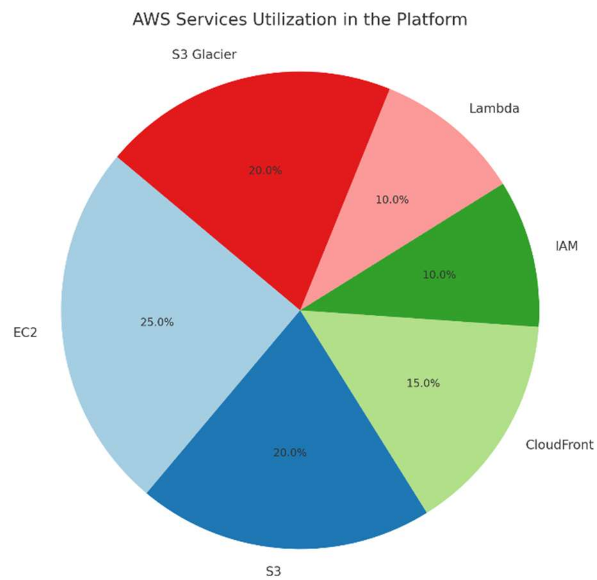

Figure 5: AWS services utilization in the platform

Figure 6: Security and availability features

6. CONCLUSION

This platform is designed to meet the increasing demand for scalable, secure, and efficient video streaming by utilizing AWS cloud services, ensuring high availability, low latency, and cost-effectiveness for media and educational applications.

The incorporation of AWS Lambda for serverless computing and AWS CloudFormation for infrastructure management will further enhance the scalability and operational efficiency of the platform.

The integration of AWS CloudFront and Auto Scaling enables the platform to dynamically scale in response to traffic fluctuations, ensuring cost efficiency, low latency, and sustained high performance.

6.1 Lessons Learned

- AWS Media Services offers a robust and scalable solution for high-performance video streaming.
- Implementing multi-AZ deployments significantly improves availability but increases operational costs.
- Using AWS IAM and AWS KMS for security and encryption is essential for protecting streaming content from unauthorized access and breaches.

6.2 Limitations

- This study assumes a fixed growth rate, whereas real-world traffic patterns can be highly variable and influenced by unpredictable factors.
- Certain AWS services may experience cost fluctuations due to dynamic pricing models, necessitating continuous cost monitoring and optimization strategies.

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8. WORKLOAD ASSIGNMENT

- Sumit K C: Research on AWS Elemental Media Services and drafting technical components. He also contributed to the analysis of CDN services

and their role in reducing latency and improving content delivery efficiency
- Khushiben Nareshbhai Patel: Writing Project Summary and Usefulness sections.
- Rajdeep Singh Golan: Assisting with research and structuring the report.
- Shila Jahanbin (Team Leader): Coordinating writing, finalizing document, disaster recovery planning, and ensuring platform scalability through AWS Auto Scaling & Load Balancing.